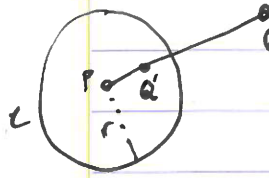
Teichmüller space and knots.

§1. Classical geometry of circles.

Defn. (e.g. Steiner, 1820s):

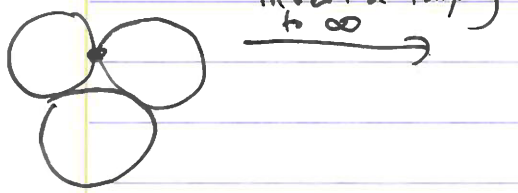


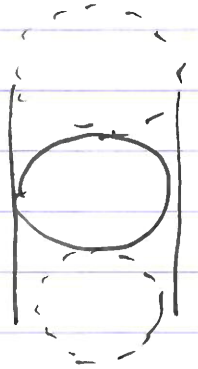
$|PQ||PQ'| = r^2$: Q and Q' are inversions ~~reflections~~ in the circle $C = C(P,r)$.

Circle inversion preserves angles & sends circles to circles. (Line = circle through ∞).
Refl. in a line is a circle inversion.

E.g. "Soddy's circles": For any 3 circles tangent to each other in a chain, $\exists!$ 2 circles mutually tangent to all three.

Proof.





In this situation, it's obvious.

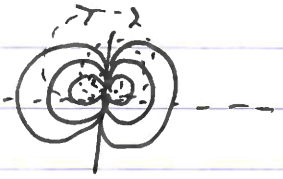
Defn. A Möbius transformation is a product of an even number of circle inversions.

Eg. 1.  Reflect in 2 parallel lines: translation.

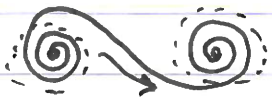
Eg. 2.  Reflect in 2 concentric circles: dilation.

Eg. 3.  Reflect in 2 intersecting circles: 'rotation'.

Thm. Every non-trivial Möbius transformation is one of three kinds:

(i) Parabolic: 

(ii) Elliptic: 

(iii) Loxodromic: 

Loxodromic = product of elliptic with

(iv) Hyperbolic: 

Thm. Möbius transformations = fractional linear maps

$$z \mapsto \frac{az+b}{cz+d} \quad (a,b,c,d \in \mathbb{C},\ ad-bc=1)$$

$$= PSL(2,\mathbb{C}) \curvearrowright \mathbb{P}^1\mathbb{C}.$$

Parabolic = $tr. = \pm 2$
elliptic = $tr \in (-2,2)$
loxodromic = $tr \notin [-2,2]$.

Generally a gp $\Gamma < PSL(2,\mathbb{C})$ has chaotic dynamics, even when Γ is discrete.

- define $\Lambda(\Gamma) = \{x \in \hat{\mathbb{C}} : x = \lim_{i\to\infty} g_i y$ for some $y \in \hat{\mathbb{C}} \setminus \{x\}$, $g_i \in \Gamma$ distinct$\}$

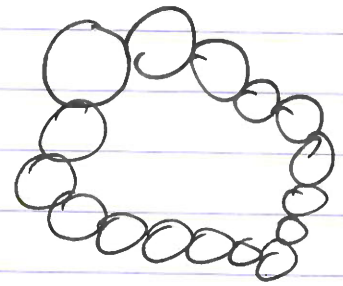
$\Omega(\Gamma) = \{x \in \hat{\mathbb{C}} :$ ~~Fan~~ open $U \ni x$ s.t. only finitely many $g \in \Gamma$ satisfy $gU \cap U \neq \emptyset\}$.

$\Lambda(\Gamma)$: "chaotic dynamics"

$\Omega(\Gamma) = \hat{\mathbb{C}} \setminus \Lambda(\Gamma)$ = "~~fa~~ lattice dynamics".

Eg. 1.

(c. 1897)



Necklace gp: generated by words of even length of reflections in a tangent chain.

Limit set $\Lambda \subset$ "set of accumulation points of orbits" is a quasicircle.

Configuration space of circles has dimension

$$3n - n - 6 \quad \leftarrow \dim \mathrm{PSL}(2,\mathbb{R})$$

↑ n = # of circles, each has center & radius.

↑ tangency conditions

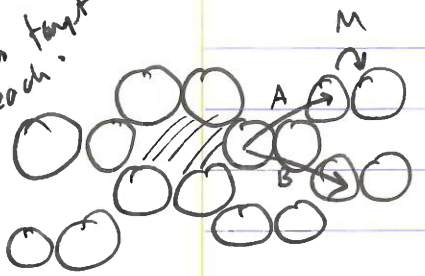
$$= 2n - 6 / \mathbb{R} = n - 3 / \mathbb{C}.$$

A natural example: Tile $\mathbb{R}^2$ with pairs of tangent circles



Parabolic.

3 circles tangent to each:

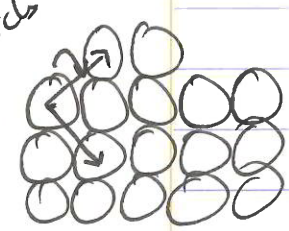


$\Gamma = \langle A, B, M \rangle$. Flexible.

$\Omega(\Gamma)/\Gamma \simeq S_{0,4}$.

↑ maximal domain in S^2 with Hausdorff quotient
= domain of discontinuity
= ~~closure of~~ complement of closure of fixed pts. of Γ.

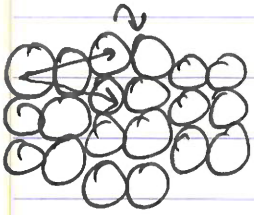
4 circles tangent to each:



quotient: $\Omega(\Gamma)/\Gamma = S_{0,5} \cup S_{0,3}$.

The corresponding groups are rigid: Small deformations are no longer discrete.

5 circles tangent to each:

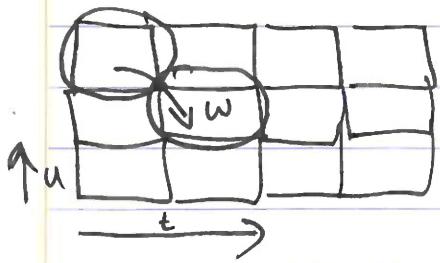


6 circles tangent to each:



No reason for the circles to be disjoint:

E.g. Neumann 1978:



$$w = \begin{pmatrix} 1 & 0 \\ -1+i & 1 \end{pmatrix} \quad t = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \quad u = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}.$$

Thm. $\langle t, u, w \rangle$ is discrete, and is a faithful representation of

$$\pi_1\left(S^3 \setminus \text{(figure-eight knot)} \right) \text{ in } PSL(2,\mathbb{C}).$$

~~**Proof.** A theorem of Riley (1975): to obtain the conclusion, it's sufficient to prove the rep^n is faithful, non-abelian, discrete,~~

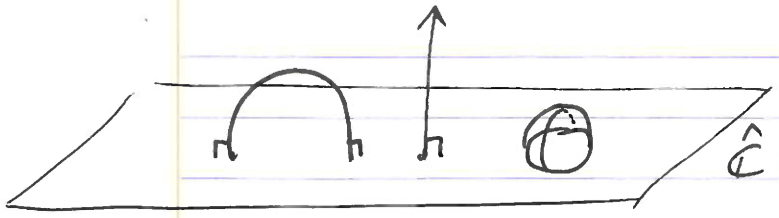
Proof: ~~A theorem of Poincaré says it is sufficient to check that sums of dihedral angles~~

How to see this?

§2. Hyperbolic geometry.

$\mathbb{H}^3 = \{(z,t) : z \in \mathbb{C},\ t > 0\}$ with the Riemannian metric

$$ds^2 = \frac{dz^2 + dt^2}{t^2}.$$



Möbius transformations act on $\mathbb{H}^3$ by extending circle inversions to inversions in hemispheres.

$\hat{\mathbb{C}} = S^2 = \partial\mathbb{H}^3$

$PSL(2,\mathbb{C}) = \mathrm{Isom}^+(\mathbb{H}^3)$.

If $\Gamma < PSL(2,\mathbb{C})$ discrete, then $\mathbb{H}^3/\Gamma$ is a hyperbolic 3-orbifold. $\Omega(\Gamma)/\Gamma = \partial_{v.c.}\ \mathbb{H}^3/\Gamma$.

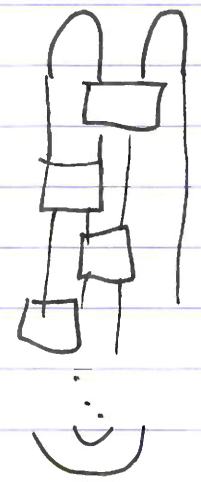
Poincaré polyhedron theorem. If a geodesic polyhedron is given in $\mathbb{H}^3$ together with isometric pairing its sides, then the isometries generate a discrete group if the polyhedron "locally" tiles, including at places when it is asymptotic to a point on the boundary. and if sides meet asymptotically with angle 0 on $\partial\mathbb{H}^3$ then the corresponding pairings are parabolic. Jan algorithm for determining a presentation of the resulting group.

Weeks's example is one application of this result.

Thurston–Riley thm: If $k \subset S^3$ is a knot which is
- not embeddable in $T^2 \subset S^3$, and is
- not a satellite, i.e. contains an embedded incomp. non-∂-parallel torus,

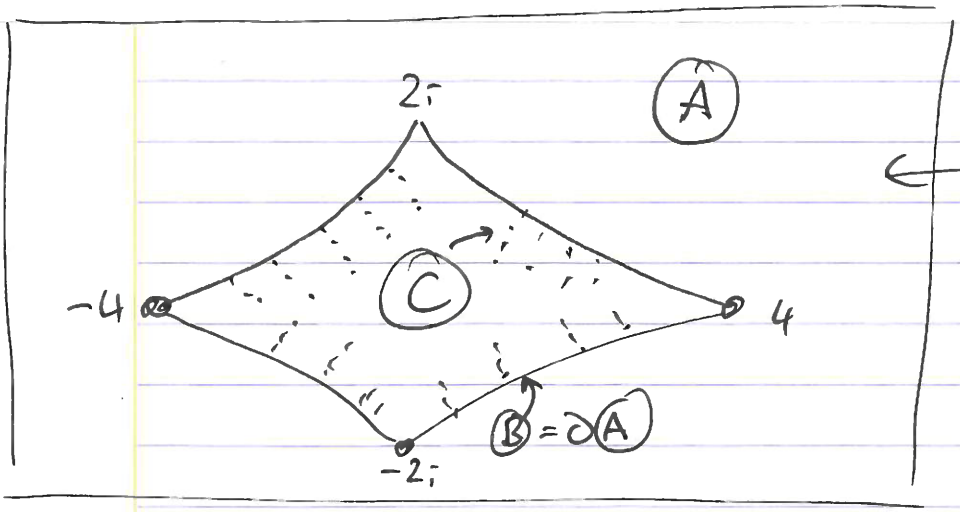
then ∃ a discrete faithful rep^n $\rho : \pi_1(S^3 \setminus k) \to PSL(2,\mathbb{C})$ and $\mathbb{H}^3/\text{im}\,\rho = S^3 \setminus k$.

Riley ('75) showed that if k is of a very special form, i.e. 2-bridge, then the image of the corresponding rep^n is of the form



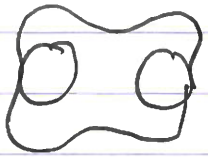
$$\Gamma_z = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix} \right\rangle.$$

Qn. For which $z \in \mathbb{C}$ is Γ_z discrete, and what are the manifolds $\mathbb{H}^3/\Gamma_z$?



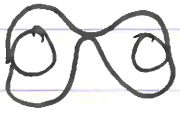
← This picture due to Riley, c. 1979.

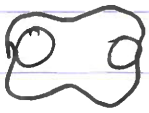
$z \in$ Region (A): Γ_z ▷ free & discrete.

$\mathbb{H}^3/\Gamma_z =$ 

$z \in$ Region (B) = ∂(A): Γ ▷ free & discrete.

2 kinds of groups:

(i) (CUSPS) $\mathbb{H}^3/\Gamma_z =$  $= 2$ $S_{0,3}$'s.

"Pinching a simple closed curve on  down to length 0".

(ii) (NON-CUSPS) $\mathbb{H}^3/\Gamma$ = "a genus 2 handlebody with no visual boundary"
= "inf. sided fund. polyhedron"
= pinching a non-cpct lamination down to length 0".

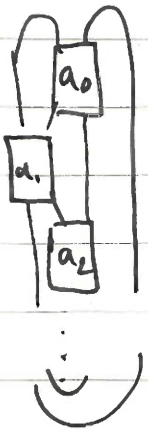
∃ a map $\mathbb{R}/\mathbb{Z} \xrightarrow{\sim}$ (B) which sends $\mathbb{Q} \to$ (i)
$\mathbb{R}\setminus\mathbb{Q} \to$ (ii).

Point ▷ lamination on $S_{0,4} \simeq \mathbb{R}/\mathbb{Z} \cup \{\infty\}$.
s.c.c's are rational numbers under this map.

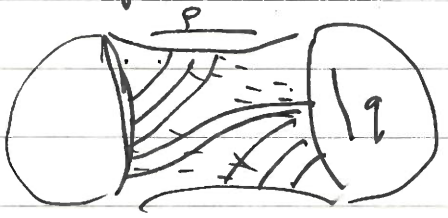
$z \in$ Region (C): For each cusp point $\in \partial$(A), take the corresponding p/q. Gives

(i) a 2-bridge link:

$$\frac{p}{q} = \cancel{a_0} + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots}}$$



(ii) a simple closed curve on $S_{0,4;\text{irr.}}$ by $W_{p/q} \in \pi_1(S_{0,4})$.



If Γ_z is discrete, then $\mathbb{H}^3/\Gamma_z$ is obtained $\overset{\text{up to finite index}}{\wedge}$ from a 2-bridge link by drilling a cone arc of angle $2\pi/n$ $(n \in [\phi, \infty])$

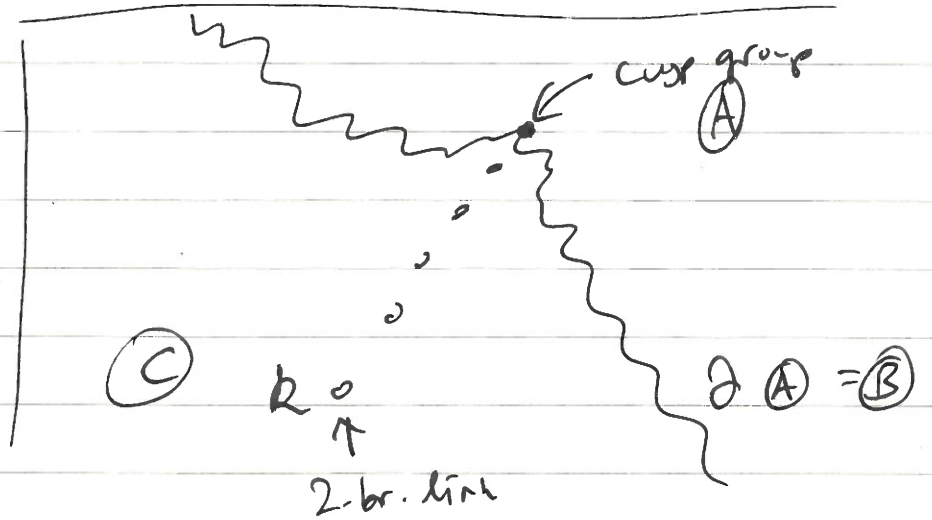
$\wedge W_{p/q}$ = elliptic of angle $2\pi/n$

(parabolic if $n = \infty$, id. if $n = \phi$).

$$\Gamma_z = \langle X, Y : \hat{W}_{p/q} = 1 \rangle.$$

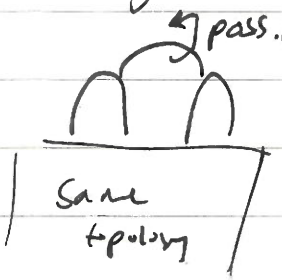
(Thm due to Akiyoshi - Ohshika × Parker - Sakuma - Yosida, '21).

* Here mention too general ellipt-ellipt



Conjectures: (Ahijoki, Sakuma, Wada, Yamashita

(i) ∃ a path 'joining the dots' consisting of a cb family of hyperbolic cone mfd holonomy groups:



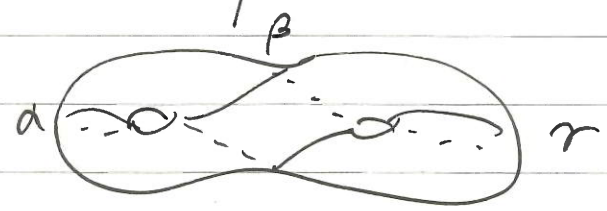
(Proved '26 by E.)

(ii) The ~~P~~ atn actually exist on the level of fundamental domains: ~~for each~~ the canonical geodesic triangulation of the link & deforms in a controlled way to give a f.dom. for the cusp group.

Essentially a "philosophy" of Riley & Sakuma.
(WIP s/w. Choudhury, Purcell.)

More general conjecture:

Fix $\mathcal{H}$ a genus 2 handlebody.



Fix 3 s.c.c. on $\partial\mathcal{H}$.

Qn. What are the discrete rep's of the form $\rho: \pi_1(\mathcal{H}) \to PSL(2,\mathbb{C})$
$\mathrm{tr}^2\rho(\alpha,\beta,\gamma) = (4,4,4)$?

- Are they all either elts. of $\overline{\mathrm{Teich}(S_{2,0})}$ or Dehn fillings of hyp. #1 links with possibly singular unknotting tunnel?
- Recent calculations (E. '26) show that the naive conjecture in this direction is false.
- Upcoming work: structure theory of hyp. #1 link groups as a first step towards understanding their canonical Δ's.